EX-NO:-14

Write a code using RAW sockets to implement packet Sniffing

AIM:-

To write a program using RAW using to capture and display network interface.

ALGORITHM:-

1) Start the program.
2) Create a RAW socket using socket(AF_PACKET, SOCK-RAW..
3) Blind the socket to the required network interfaces.
4) Extract the ethernet header from each packets.
5) Display the packet information.
6) Repeat unit the user terminates the program.
7) Close the socket.

PROGRAM:-

```
import socket
import struct
# create a RAW socket
raw_socket = socket.socket(
    socket.SOCK_RAW,
    socket.ntohs(3)
)
print("Packing Sniffer started...")
print("Waiting for packets..\n")
try:
    while True:
    packet, address = raw_socket.recvfrom(65535)
 # Ethernet header is first 14 bytes
 ethernet_header = packet[:14]
 # unpack ethernet header
  '!6s6sH', ethernet_header
 )
 # Convert MAC addresses
 ":".02x".f. byte for byte in dest_mac
)
print("=========")
print("source MAC:", src_mac)
print("Protocol:", hex(Protocol)
```

OUTPUT :-

Packet Sniffer started..

Waiting for packets..

Source MAC : 3C:52:82:11:22:33

Destination MAC : ff-ff-ff:ff:ff:ff

Protocol : 0X806

packet type : ARP

Source MAC : 3C:52:82:11:22:33

Source IP : 192.168.1.10

Destination IP : 192.168.1.1

TTL : 64

IP Protocol : 6

```
print ("Packet Type : ARP")
else:
    Print ("Packet Type : other")
except keyboardinterrupt :
 print ("\n Packet sniffing stopped.")
 finally:
    oooo_socket.close ()
```

Sample output :-

packet sniffer started.

Source MAC : 3C:52:82:11:22:33
Destination MAC : ff:ff:ff:ff:ff:ff
Protocol : 0x806
Packet Type : ARP

Source MAC : 3C:52:82:11:22:33
Source IP : 192.168.1.10
Destination IP : 192.168.1.1
TTL : 64
IP Protocol : 6

RESULT :-

Thus, a packet-sniffing program using RAW sockets was successfully implemented. The program capture network packets and displays important information such as Source MAC address, destination MAC address, protocal, Source IP address, destination IP address, and TTL.